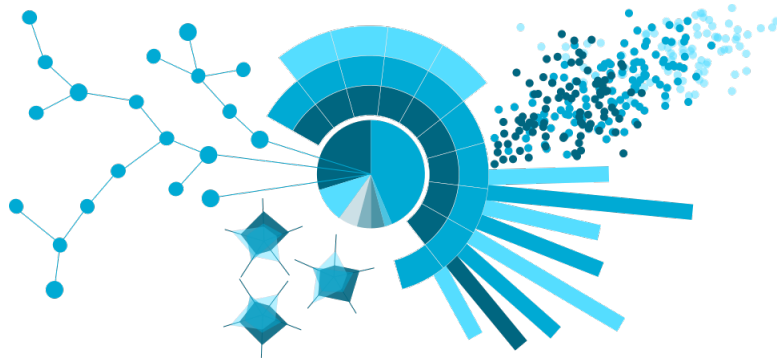


Data Visualization
Assignment 1
Group 4



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November 15, 2025

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1 Introduction

The assignment was to answer 3 sets of questions with visualizations and observations for the specified tasks using Jupyter Notebook. Both Preprocessing decisions and insights are summarized for each section.

2 Q1–3

For the first tasks, the CSV datasets were loaded with the `transformed` date column to datetimes and set as the index. Data were filtered to the range 1992–2002 (Irrelevant dates were removed). For comparison clarity, monthly averages were computed for the third plot.

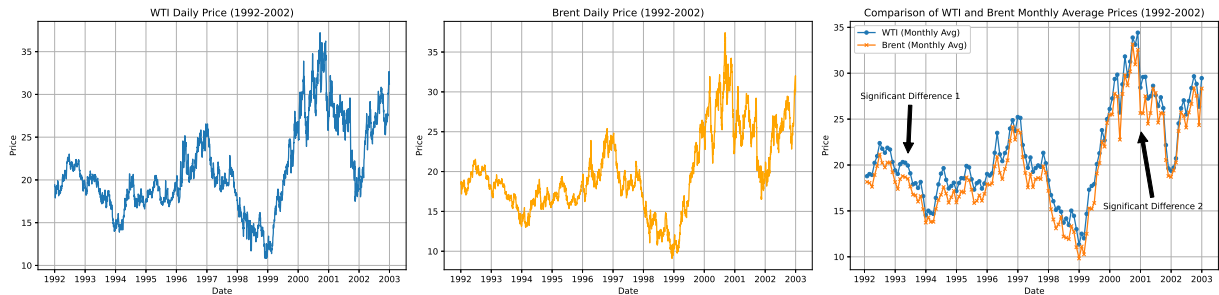


Figure 1: WTI vs Brent oil prices: daily (WTI, Brent) and monthly average comparison (1992–2002).

Key observations:

- Significant differences in the third task were decided using early versions of the plot and human judgment. Another approach would be to write a function that checks when a certain difference between the 2 prices exists, and use this as the basis for the markers.
- The different colors for the lines were crucial to compare the two in the third plot, but the versatility of the different marks was less obvious.
- The task, while fairly straightforward, mostly turned into a workshop for Gemini and how to balance the use of automatic tools against manual programming.

3 Q4–5

The Sales data, originally obtained from an Excel file, was cleaned and prepared prior to analysis. This included handling missing values, ensuring consistent data types, and removing duplicates. After preprocessing, the data was grouped to calculate total sales by city and payment method. These collected values were used to prepare bar charts for comparison. The following figure shows both distributions and annotates the maxima.

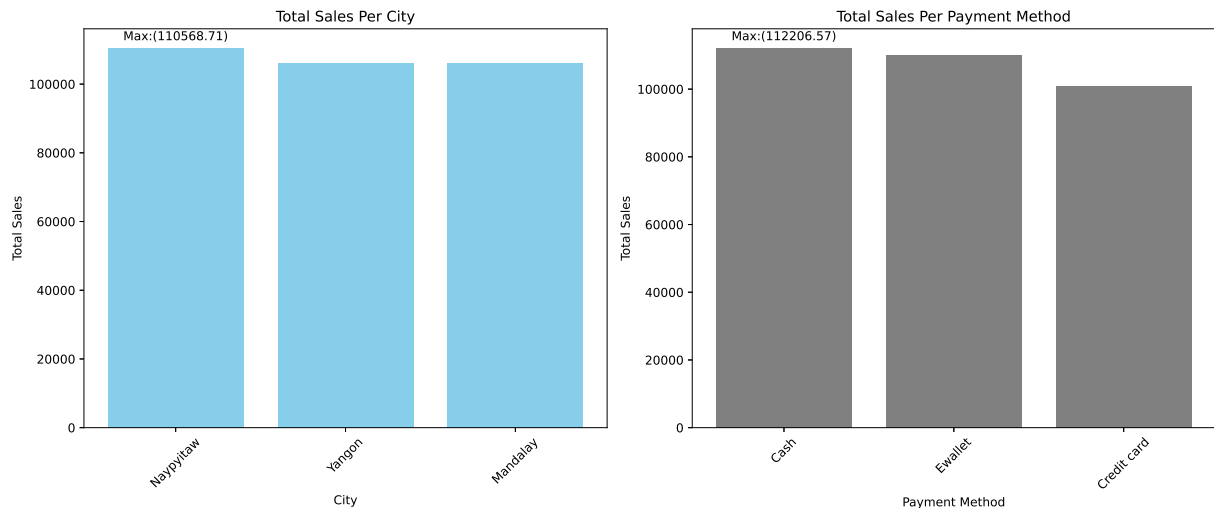


Figure 2: City-wise total sales (left) and payment-method total sales (right).

Key observations:

1. **City-wise Sales Distribution.** From the plotted (left side) bar chart of city-wise sales distribution, the total sales for each city are shown. The values are:
 - **Naypyitaw:** ~110,568
 - **Yangon:** ~107,000
 - **Mandalay:** ~107,000So, **Naypyitaw** has contributed the maximum sales, with a total of **110,568.71**.
2. **Payment Method Influence on Total Sales:** From the bar chart plotted (on the right), it is clear that payment methods affect total sales, as each method contributes a different amount to the overall revenue. Total sales vary across three payment types: **Cash**, **Ewallet** and **Credit card**. The noticeable difference in the height of the bars clearly indicates that customers' choice of payment method affects the total revenue collected.
3. **Payment method with highest contribution:** **Cash** has the highest total sales with **112,206.57**. This shows that **Cash** is the most significant contributor to total sales, followed by **Ewallets**, while **Credit cards** are the least used payment method in terms of revenue generated.

4 Q6–7

For the last set of tasks the image was loaded using matplotlib as a numpy array and visualized using imshow, then the photo was transformed to grayscale using rgb2gray and the cmaps were applied in a 2*3 grid for a total of 6 visualizations. The cmaps used are:(gray, viridis, magma, hot, coolwarm, hsv)

For Q7, the raw image was loaded and filters such as sharpen, gaussian_blur, contrast enhance, brightness enhance, edge detection, grayscale were applied. The image was then loaded into a proprietary grid.

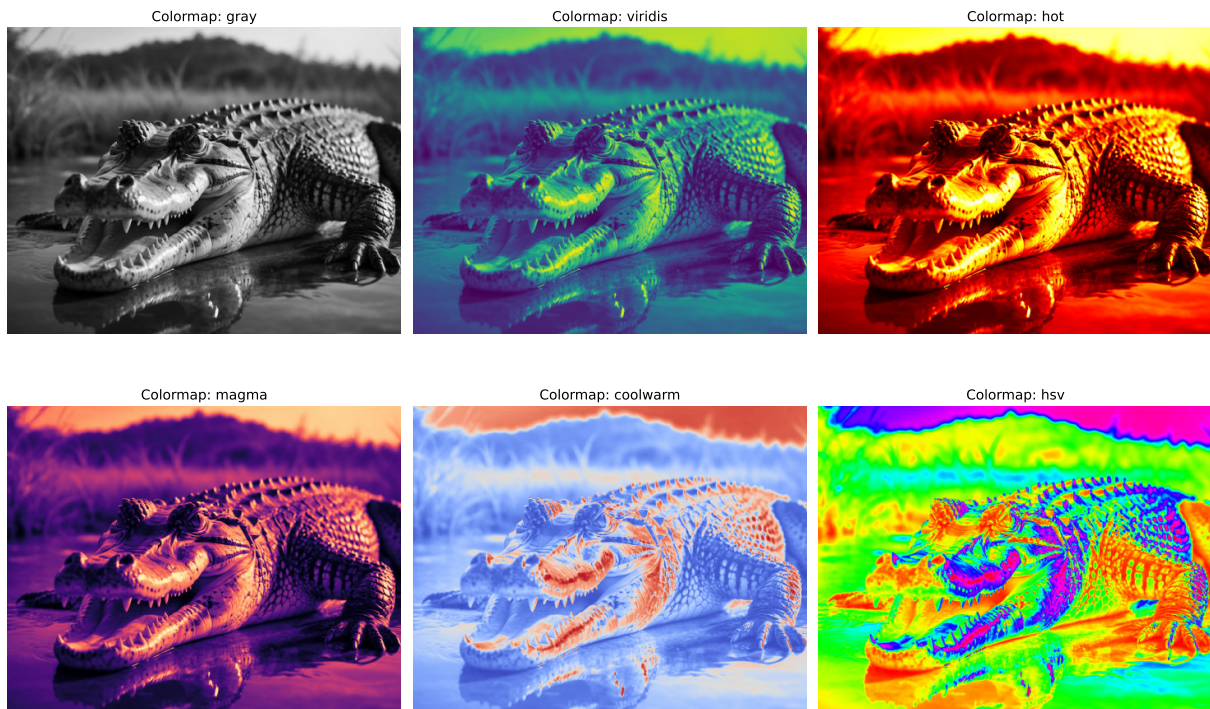


Figure 3: Six colormaps applied to the grayscale image (gray, viridis, hot, magma, coolwarm, hsv).



Figure 4: Six filters applied to the image in a custom grid layout.

5 Final Reflections

- The assignment was mostly done in parallel, meaning that while the tasks were carried out quickly, the solutions lacked unity. For the next assignment spending more time on making this a team effort will likely increase how much is learnt, as it will force the group members to continuously articulate what they are doing.
- The two latter tasks, q4-5 and q6-7, were done twice. The results were similar but not identical.
- In the third task, q6-7, it was noted that `cmaps` does not apply to RGB images but get overruled, thus to use `cmaps` a single channel(Grayscale) image is needed.

6 Group Members Respective Contributions

- Ahmed Selim Al Baker Khan: q4-5
- Ashraf Sarhan: q4-5, q6-7, Latex
- Egil Elenius: q1-3, report writing
- Zeyad Haroun: q6-7, notebook structuring
- Abdalrahman Dabor